University of Florida, Department of Mathematics

Analysis First-Year Exam, May 2023, Part 2

Answer FOUR questions in detail. State carefully any results used without proof.

1. Prove that the function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable if and only if for each $\epsilon > 0$ there exist Riemann-integrable functions ℓ and u on $[a, b]$ such that $\ell \leq f \leq u$ and $\int_a^b (u - \ell) < \epsilon$.

2. Let $f : [0, 1] \to \mathbb{R}$ be continuous; for each $n \in \mathbb{N}$ and each $t \in [0, 1]$ define $f_n(t) = f(t)t^n$. Prove that the sequence $(f_n : n \in \mathbb{N})$ converges uniformly on $[0, 1]$ if and only if $f(1)$ has a specific value, which should be found.

3. Let $f : [-1, 1] \to \mathbb{R}$ be continuous. Prove that if

$$\int_{-1}^{1} f(t)t^n \, dt = 0$$

whenever the natural number n is odd, then the function f is even.

4. Let $(f_n : n \in \mathbb{N})$ be a sequence of measurable real-valued functions on some measurable space. Prove that each of these three sets is measurable:

$$\begin{aligned} C &= \{\omega : \text{ the sequence } f_n(\omega) \text{ is Cauchy}\}; \\ D &= \{\omega : \text{ the sequence } f_n(\omega) \text{ has all terms different}\}; \\ E &= \{\omega : \text{ the sequence } f_n(\omega) \text{ is eventually constant}\}. \end{aligned}$$

5. The bounded function $f : [0, 1] \times [0, 1] \to \mathbb{R}$ is separately continuous: that is, $f(x, y)$ is continuous in x when y is fixed and continuous in y when x is fixed. Prove the continuity of the function $F : [0, 1] \to \mathbb{R}$ defined by

$$F(x) = \int_0^1 f(x, y) \, dy.$$